Class XII Session 2025-26
Subject - Mathematics
Sample Question Paper - 4

Time Allowed: 3 hours

General Instructions:

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A,
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. Use of calculators is not allowed.

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Section A

1. Let A = {1, 2, 3} and let R = {(1,1), (2, 2), (3, 3), (1, 2), (2,1), (2, 3), (3, 2)}. Then, R is **[1]**

   a) reflexive and symmetric but not transitive
   b) an equivalence relation
   c) symmetric and transitive but not reflexive
   d) reflexive and transitive but not symmetric

2. The value of $\sin^{-1}(\cos\frac{\pi}{9})$ is **[1]**

   a) $\frac{5\pi}{9}$
   b) $\frac{7\pi}{18}$
   c) $\frac{-5\pi}{9}$
   d) $\frac{\pi}{9}$

3. For which value of x, are the determinants $\begin{vmatrix} 2x & -3 \\ 5 & x \end{vmatrix}$ and $\begin{vmatrix} 10 & 1 \\ -3 & 2 \end{vmatrix}$ equal? **[1]**

   a) ± 2
   b) 2
   c) ± 3
   d) -3

4. The function f(x) = |cos x| is **[1]**

   a) everywhere continuous but not differentiable at $(2n + 1)\frac{\pi}{2}$, $n \in Z$
   b) either continuous or differentiable at $(2n + 1)\frac{\pi}{2}$, $n \in Z$
   c) neither continuous nor differentiable at $(2n + 1)\frac{\pi}{2}$, $n \in Z$
   d) everywhere continuous and differentiable

5. If a vector makes an angle of $\frac{\pi}{4}$ with the positive directions of both x-axis and y-axis, then the angle which it **[1]**

makes with positive z-axis is:

a) 0 b) $\frac{3\pi}{4}$

c) $\frac{\pi}{2}$ d) $\frac{\pi}{4}$

6. Consider the following statements in respect of the differential equation $\frac{d^2y}{dx^2} + \cos\left(\frac{dy}{dx}\right) = 0$ **[1]**

i. The degree of the differential equation is not defined.

ii. The order of the differential equation is 2. Which of the above statement(s) is/are correct?

a) Neither (i) nor (ii) b) Both (i) and (ii)

c) Only (ii) d) Only (i)

7. The graph of the inequality 2x + 3y > 6 is **[1]**

a) whole XOY – plane excluding the points on the line 2x + 3y = 6.

b) half plane that neither contains the origin nor the points of the line 2x + 3y = 6.

c) entire XOY plane.

d) half plane that contains the origin.

8. Magnitude of the vector $\vec{a} = 2i - 7j - 3k$ is **[1]**

a) $\sqrt{61}$ b) $\sqrt{65}$

c) $\sqrt{62}$ d) $\sqrt{63}$

9. $\int \frac{dx}{(2-3x)}$ =? **[1]**

a) - 3 log |2 – 3x| + C b) log |2 + 3x| + C

c) - log |2 – 3x| + C d) $-\frac{1}{3}\log|2-3x| + C$

10. The order of matrix A = $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & -1 \end{bmatrix}$ is _____. **[1]**

a) 2 × 3 b) 3 × 3

c) 2 × 2 d) 3 × 2

11. The point at which the maximum value of x + y, subject to the constraints x + 2y ≤ 70, 2x + y ≤ 95, x, y ≥ 0 is obtained, is **[1]**

a) (30, 25) b) (20, 35)

c) (35, 20) d) (40,15)

12. For what value of λ, the projection of vector $i + \lambda j$ on vector $i - j$ is $\sqrt{2}$? **[1]**

a) 0 b) -1

c) 1 d) 3

13. $\begin{vmatrix} \cos 70^\circ & \sin 20^\circ \\ \sin 70^\circ & \cos 20^\circ \end{vmatrix} = ?$ **[1]**

a) $\sin 50^o$ b) 0

c) $\cos 50^o$ d) 1

14. A bag contains 5 red and 3 blue balls. If 3 balls are drawn at random without replacement the the probability that exactly two of the three balls were red, the first ball being red, is

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a) $\frac{4}{7}$

b) $\frac{5}{8}$

c) $\frac{15}{28}$

d) $\frac{1}{3}$

15. General solution of y log y dx – x dy = 0 **[1]**

a) $y = e^{cx}$

b) $y^2 = e^{cx}$

c) $y = e^{cx} + e^{-cx}$

d) $y = e^{-cx}$

16. Let L denote the set of all straight lines in a plane. Let a relation R be defined by lRm if and only if l is perpendicular to m ∀ l, m ∈ L. Then R is **[1]**

a) reflexive

b) symmetric

c) Asymmetric

d) transitive

17. For what value of k may the function $f(x) = \begin{cases} k(3x^2 - 5x), & x \le 0 \\ \cos x, & x > 0 \end{cases}$ become continuous? **[1]**

a) 1

b) $-\frac{1}{2}$

c) 0

d) No value

18. The straight line $\frac{x-3}{3} = \frac{y-2}{1} = \frac{z-1}{0}$ is **[1]**

a) perpendicular to z-axis

b) parallel to x-axis

c) parallel to y-axis

d) parallel to z-axis

19. **Assertion (A):** The rate of change of area of a circle with respect to its radius r when r = 6 cm is 12π cm^2/cm. **[1]**

Reason (R): Rate of change of area of a circle with respect to its radius r is $\frac{dA}{dr}$, where A is the area of the circle.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

20. The value of λ for which the two vectors $2i - j + 2k$ and $3i + \lambda j + k$ are perpendicular is **[1]**

a) 8

b) 4

c) 6

d) 2

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